



Department of Electronic Engineering

Industrial Electronics

Course Code: EL-301

Open Ended Lab Report

Group Members	Roll Numbers
Minahil Shafique	EL-043
Aafia Shah	EL-058
Abdullah Khan	EL-064
Muhammad Asad Sajid	EL-068

Instructor: Ms. Hira Imtiaz

Semester: 5th (Fall-2024), TE

Report: PLC-Based Car Parking Control System

1. Introduction

The objective of this project is to design a PLC-based car parking control system for a shopping mall, managing up to 500 vehicles per floor across four parking levels: Lower Ground (LG), Basement One (B1), Basement Two (B2), and Basement Three (B3). Each floor's parking control system will be integrated and managed by a central PLC system using ladder logic programming.

The PLC system for this project will be implemented on a SIMATIC S7-300 series PLC, and the logic will handle parking availability, full conditions, and system activation and deactivation using input/output devices such as push buttons, LEDs, and sensors.

2. System Overview

The parking system is designed to operate on four levels:

1. **Lower Ground (LG)** - Valet Parking
2. **Basement One (B1)** - Self-Parking
3. **Basement Two (B2)** - Self-Parking
4. **Basement Three (B3)** - Self-Parking

The system will:

- Track the number of vehicles on each floor.
- Indicate the availability of parking space.
- Manage vehicle entry/exit based on parking capacity.
- Display parking full indicators when capacity is reached.

3.Memory Address Assignment for Contacts, Coils, and LEDs

Input/Output Device	Function	PLC Address
Parking System Enable PB	Enable parking system	I0.0
Parking System Disable PB	Disable parking system	I0.1
LG Parking Full Indicator	LED for LG full	Q0.3
B1 Parking Full Indicator	LED for B1 full	Q0.6
B2 Parking Full Indicator	LED for B2 full	Q1.1
B3 Parking Full Indicator	LED for B3 full	Q1.3
No Parking Available LED	LED for No Parking available	Q1.5
LG Entry Sensor	Car enters LG	I0.2
LG Exit Sensor	Car exits LG	I0.3
B1 Entry Sensor	Car enters B1	I0.4
B1 Exit Sensor	Car exits B1	I0.5
B2 Entry Sensor	Car enters B2	I0.6
B2 Exit Sensor	Car exits B2	I0.7
B3 Entry Sensor	Car enters B3	I0.6
B3 Exit Sensor	Car exits B3	I0.7

4. Ladder Networks

4.1 System Activation and Deactivation

Description: The parking system is activated when the "Parking System Enable" push button is pressed and deactivated when the "Parking System Disable" button is pressed. LEDs indicate the system status (enabled or disabled).

Ladder Logic:

```
| I0.0 |----| Parking System Enable |----- ( S )----| Q0.0 (Enable LED) |  
| I0.1 |----| Parking System Disable |----- ( R )----| Q0.1 (Disable LED) |
```

Explanation:

- **I0.0** (Enable PB) sets the system to activate and turns on Q0.0 (Enable LED).
- **I0.1** (Disable PB) resets the system and turns off the enable state, turning off Q0.0.

4.2 Valet Parking (LG)

Description: The LG floor is dedicated to valet parking. A car is counted upon entry, and the floor's available parking is tracked. When the count reaches 500, no further cars are allowed, and the "LG Parking Full" LED is turned on.

Ladder Logic:

```
| I1.0 |----| LG Entry Sensor |----[ Counter Up C0 ]---|  
| I1.1 |----| LG Exit Sensor |----[ Counter Down C0 ]---|  
| C0 = 500 |----( Q0.0 )----| LG Full Indicator |
```

Explanation:

- **I1.0** and **I1.1** represent the car entry and exit sensors, respectively.
- **C0** counts the cars on the LG floor. When it reaches 500, **Q0.0** (LG Full LED) is turned on.

4.3 Self-Parking Basement One (B1)

Description: B1 is the first level for self-parking. If there are fewer than 500 cars, a new car can park, and the counter increases. When the parking is full, the "B1 Full" LED is turned on.

Ladder Logic:

```
| I1.2 |----| B1 Entry Sensor |----[ Counter Up C1 ]---|  
| I1.3 |----| B1 Exit Sensor  |----[ Counter Down C1 ]---|  
| C1 = 500 |----( Q0.1 )----| B1 Full Indicator |
```

Explanation:

- **C1** keeps track of cars on the B1 floor, and **Q0.1** is activated when B1 reaches 500 cars.

4.4 Self-Parking Basement Two (B2)

Description: If B1 parking is full, vehicles are redirected to B2. The same logic applies here, and the "B2 Full" LED is lit when the floor reaches capacity.

Ladder Logic:

```
| I1.4 |----| B2 Entry Sensor |----[ Counter Up C2 ]---|  
| I1.5 |----| B2 Exit Sensor  |----[ Counter Down C2 ]---|  
| C2 = 500 |----( Q0.2 )----| B2 Full Indicator |
```

Explanation:

- **C2** counts cars on B2, and **Q0.2** (B2 Full LED) turns on when 500 cars are parked.

4.5 Self-Parking Basement Three (B3)

Description: B3 is the last level for self-parking. When both B1 and B2 are full, vehicles are allowed to park in B3. Once full, the "B3 Full" LED lights up.

Ladder Logic:

```
| I1.6 |----| B3 Entry Sensor |----[ Counter Up C3 ]---|  
| I1.7 |----| B3 Exit Sensor  |----[ Counter Down C3 ]---|  
| C3 = 500 |----( Q0.3 )----| B3 Full Indicator |
```

Explanation:

- **C3** tracks cars in B3, and **Q0.3** lights up when B3 parking is full.

4.6 All Parking Full (No More Parking Allowed)

Description: When all floors (LG, B1, B2, B3) are full, the system must prevent further parking. The "No Parking Available" LED is activated.

Ladder Logic:

```
| C0 = 500 |----| C1 = 500 |----| C2 = 500 |----| C3 = 500 |----  
(Q0.4)----| No Parking Indicator |
```

Explanation:

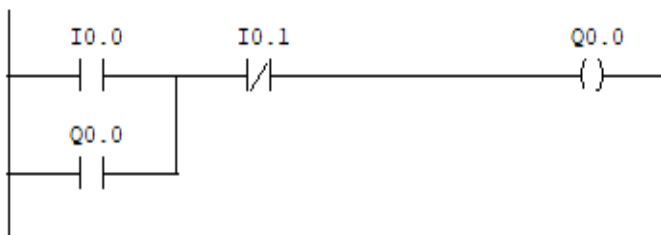
- When all floors are full (C0, C1, C2, C3), the "No Parking Available" LED (**Q0.4**) is turned on.

5. Snapshots of Behavior under Different Scenarios

OB1 : "Main Program Sweep (Cycle)"

```
system:
system on---input1=I0.0
system of---input2=I0.1
system output---Q0.0
system on led---Q0.1
system off led---Q1.7
GROUND FLOOR:
car entering sensor---input3=I0.2
car leaving sensor---input4=I0.3
for ground floor counter---MW0=for storing counter values
ground floor parking full output---Q1.2
to limit the counter---output---Q2.0
led indicating that ground floor is full---Q0.3
led indicating that ground floor is not full---Q0.4
FIRST FLOOR:
car entering sensor---input5=I0.4
car leaving sensor---input6=I0.5
for first floor counter---MW1 for storing counter values
to limit the counter---output---Q2.1
first floor parking full output---Q0.5
```

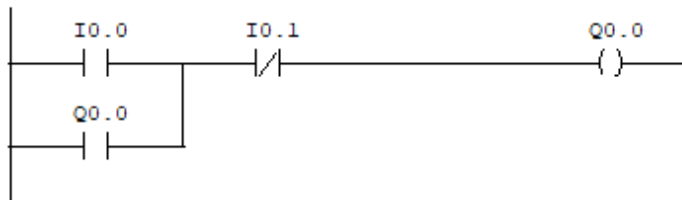
Network 1: SYSTEM



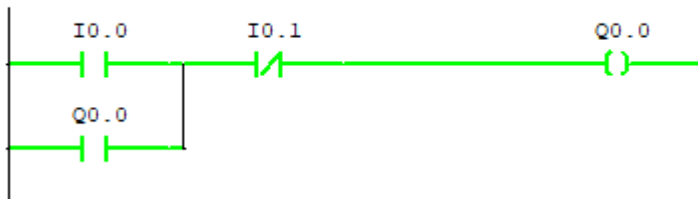
OB1 : "Main Program Sweep (Cycle)"

```
to limit the counter---output---Q2.1
first floor parking full output----Q0.5
led indicating that first floor is full---Q0.6
led indicating that first floor is not full---Q0.7
SECOND FLOOR:
car entering sensor---input5=I0.6
car leaving sensor---input6=I0.7
for second floor counter---MW2 for storing counter values
to limit the counter---output---Q2.2
second parking full output----Q1.0
led indicating that second is full---Q1.1
led indicating that second is not full---Q1.2
THIRD FLOOR:
car entering sensor---input5=I0.6
car leaving sensor---input6=I0.7
for third floor counter---MW3 for storing counter values
to limit the counter---output---Q2.3
second parking full output----Q1.6
led indicating that second is full---Q1.3
led indicating that second is not full---Q1.4
```

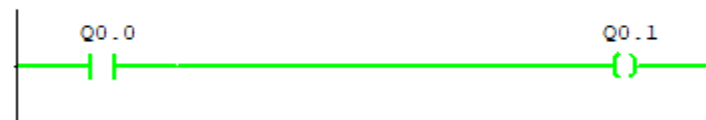
Network 1: SYSTEM



Network 1: SYSTEM



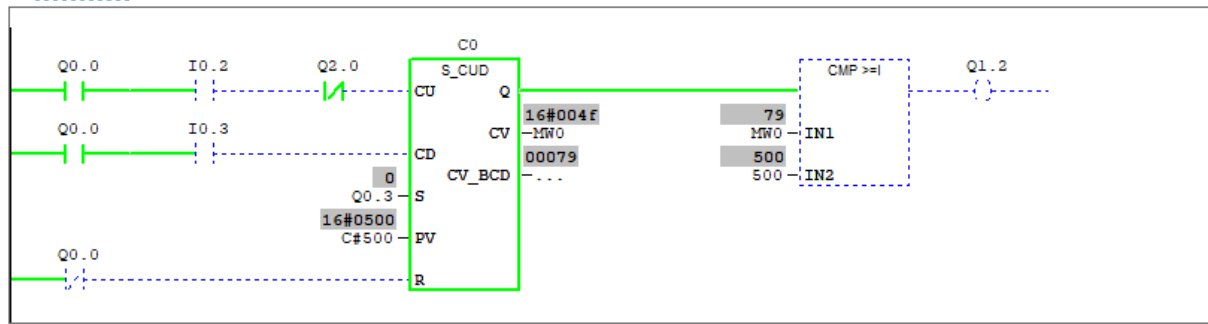
Network 2 : SYSTEM ON LED



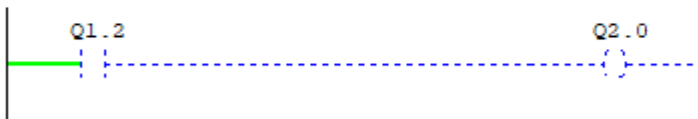
Network 3 : SYSTEM OFF LED



Network 4: COUNTER FOR GROUND FLOOR



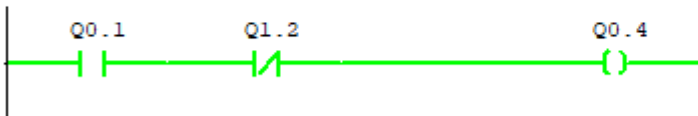
Network 5: TO LIMIT THE CU FOR GROUND FLOOR



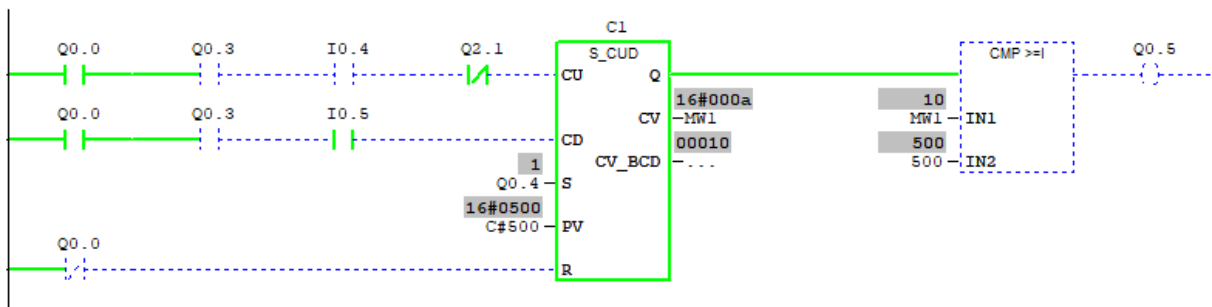
Network 6: LED ON WHEN PARKING IS FULL FOR GOUND FLOOR



Network 7: LED ON WHEN PARKING IS NOT FULL FOR GOUND FLOOR



Network 8: COUNTER FOR FIRST FLOOR



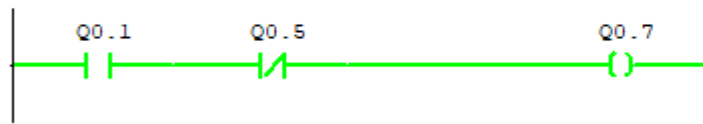
Network 9: TO LIMIT THE CU FOR FIRST FLOOR



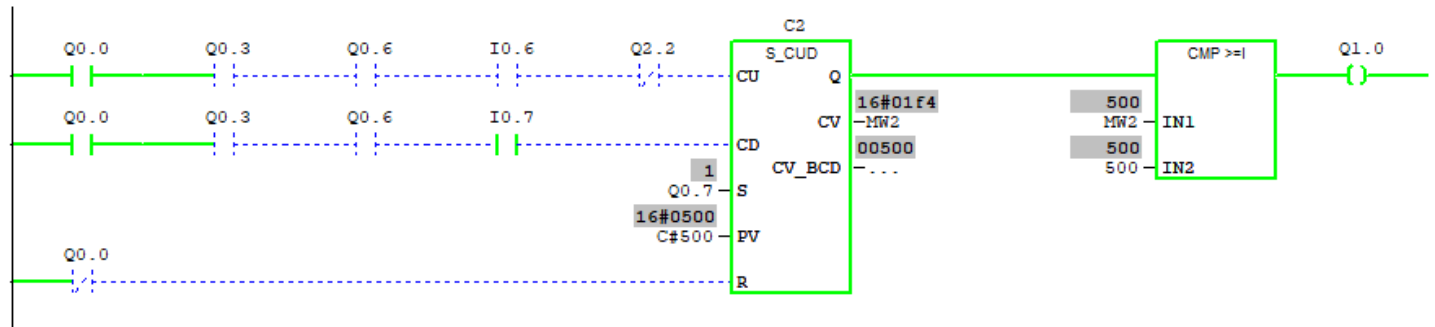
Network 10: LED ON WHEN PARKING IS FULL FOR FIRST FLOOR



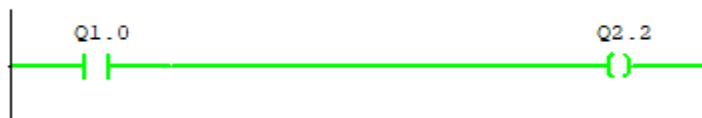
Network 11 : LED ON WHEN PARKING IS NOT FULL FOR FIRST FLOOR



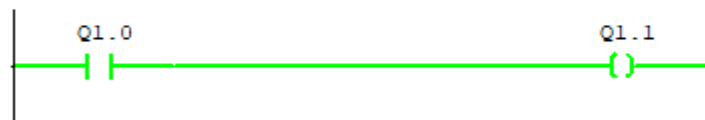
Network 12 : COUNTER FOR SECOND FLOOR



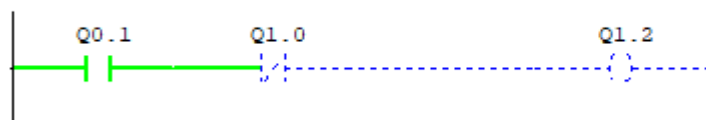
Network 13 : TO LIMIT THE CU FOR SECOND FLOOR



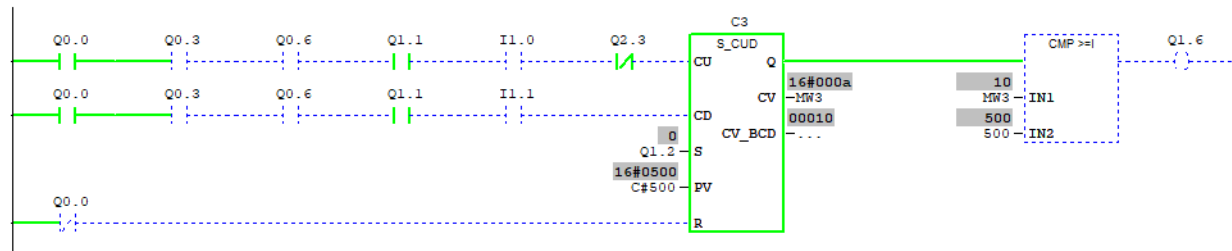
Network 14 : LED ON WHEN PARKING IS FULL FOR SECOND FLOOR



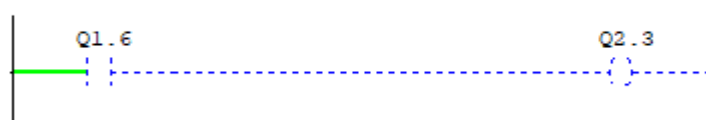
Network 15 : LED ON WHEN PARKING IS NOT FULL FOR SECOND FLOOR



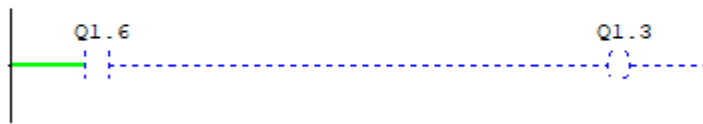
Network 16 : COUNTER FOR THIRD FLOOR



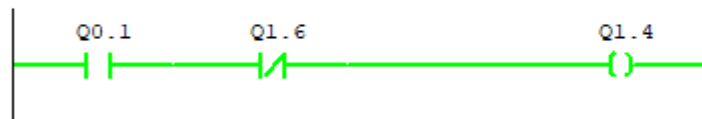
Network 17 : TO LIMIT THE CU FOR THIRD FLOOR



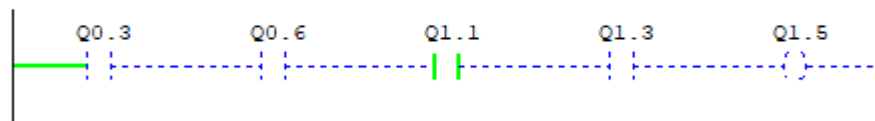
□ Network 18 : LED ON WHEN PARKING IS FULL FOR THIRD FLOOR



□ Network 19 : LED ON WHEN PARKING IS NOT FULL FOR THIRD FLOOR



□ Network 20 : LED ON WHEN EACH FLOOR PARKING IS FULL

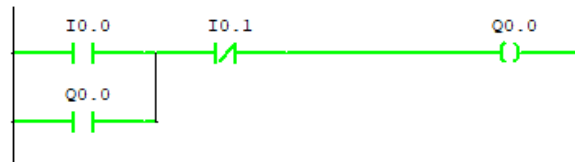


OBI : "Main Program Sweep (Cycle)"

```

system:
system on---input1=I0.0
system of---input2=I0.1
system output---Q0.0
system on led---Q0.1
system off led---Q1.7
GROUND FLOOR:
car entering sensor---input3=I0.2
car leaving sensor---input4=I0.3
for ground floor counter---MW0=for storing counter values
ground floor parking full output---Q1.2
to limit the counter---output---Q2.0
led indicating that ground floor is full---Q0.3
led indicating that ground floor is not full---Q0.4
FIRST FLOOR:
car entering sensor---input5=I0.4
car leaving sensor---input6=I0.5
for first floor counter---MW1 for storing counter values
to limit the counter---output---Q2.1
first floor parking full output---Q0.5
  
```

□ Network 1 : SYSTEM



□ Network 2 : SYSTEM ON LED

S7-PLCSIM1

File Edit View Insert PLC Execute Tools Window Help

PLCSIM(MPI)

IB 0 Bits

7	6	5	4	3	2	1	0
☒	☒	☒	☐	☐	☐	☐	☒

IB 1 Bits

7	6	5	4	3	2	1	0
☐	☐	☐	☐	☐	☐	☐	☐

QB 1 Bits

7	6	5	4	3	2	1	0
☐	☐	☒	☐	☐	☐	☒	☒

QB 0 Bits

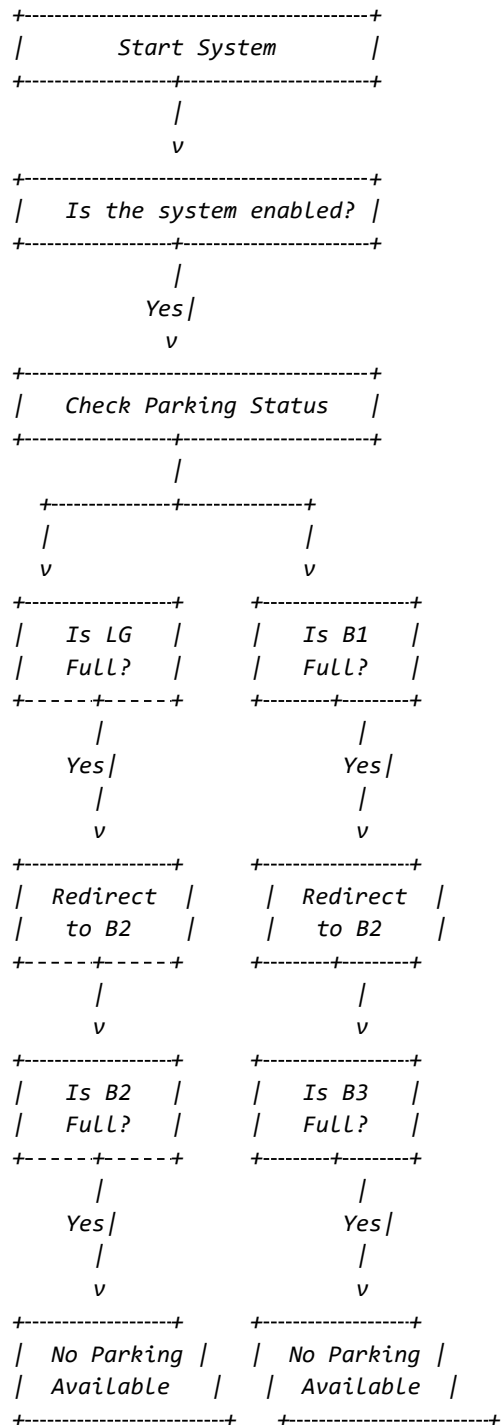
7	6	5	4	3	2	1	0
☒	☐	☐	☒	☐	☐	☒	☒

Press F1 to get Help.

Scenario	Expected Behavior
System Enabled	Parking floors are activated, and LEDs indicate parking availability.
LG Parking Full	No more cars allowed on LG, "LG Full" LED turned on.
B1 Parking Full	No more cars allowed on B1, redirected to B2, "B1 Full" LED on.
All Parking Full	"No Parking Available" LED turned on, no further entry.

6. Pictorial Representation of the Parking System

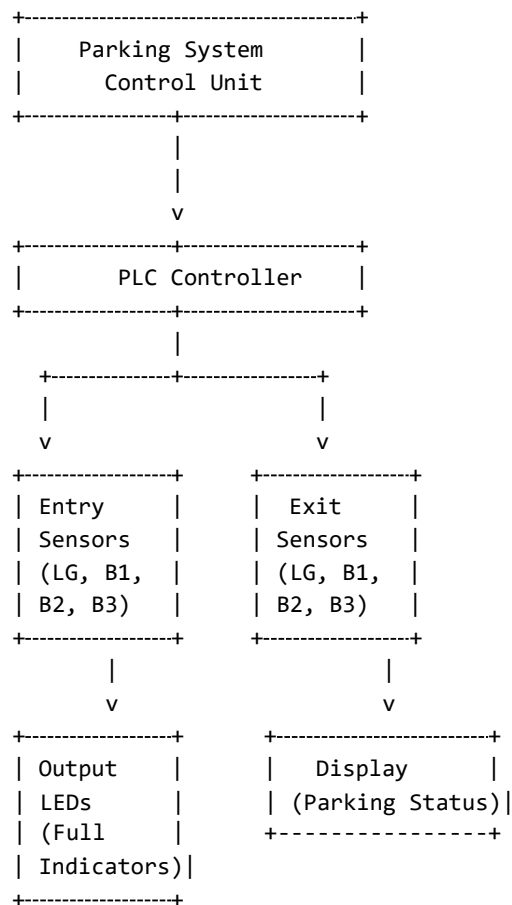
6.1 Flowchart of the Parking System Logic



Explanation:

- The flowchart starts by determining if the parking system is enabled.
- If enabled, it checks the parking status for each level (LG, B1, B2, B3) in sequence.
- Vehicles are redirected to the next available level if the current one is full.
- If all levels are full, a "No Parking Available" status is indicated.

6.2 Block Diagram of the Parking Control System



Explanation:

- The block diagram illustrates the major components of the parking control system.
- It includes entry and exit sensors for each level (LG, B1, B2, B3) that monitor vehicle presence.
- The PLC controller processes data from these sensors.
- Output LEDs indicate whether a parking level is full, while a display module provides real-time status of the parking system.

7. Softcopy of the Program

The softcopy of the program is attached, and it can be tested on the SIMATIC S7-300 PLC for different input conditions.

8. Conclusion

This report outlines the design and functionality of a PLC-based car parking control system that manages the parking for 500 vehicles per floor across multiple parking levels. The system is designed to efficiently manage parking availability, indicating full or available conditions using LEDs and managing car counts for each floor.